

Customer Shopping Behavior Analysis

A comprehensive data-driven exploration of **3,900 transactions** across product categories — uncovering spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

DATA SCIENCE

PYTHON · SQL · POWER BI



Made with GAMMA

Project Overview & Dataset Summary

This project analyzes customer shopping behavior using transactional data from **3,900 purchases** across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions. The dataset comprises **18 columns** and covers three broad feature groups: customer demographics, purchase details, and shopping behavior signals.

Dataset at a Glance

3,900

Total transaction rows

18

Feature columns

37

Missing values in Review Rating

Key Feature Groups

Customer Demographics: Age, Gender, Location, and Subscription Status provide foundational segmentation attributes for understanding who is purchasing.

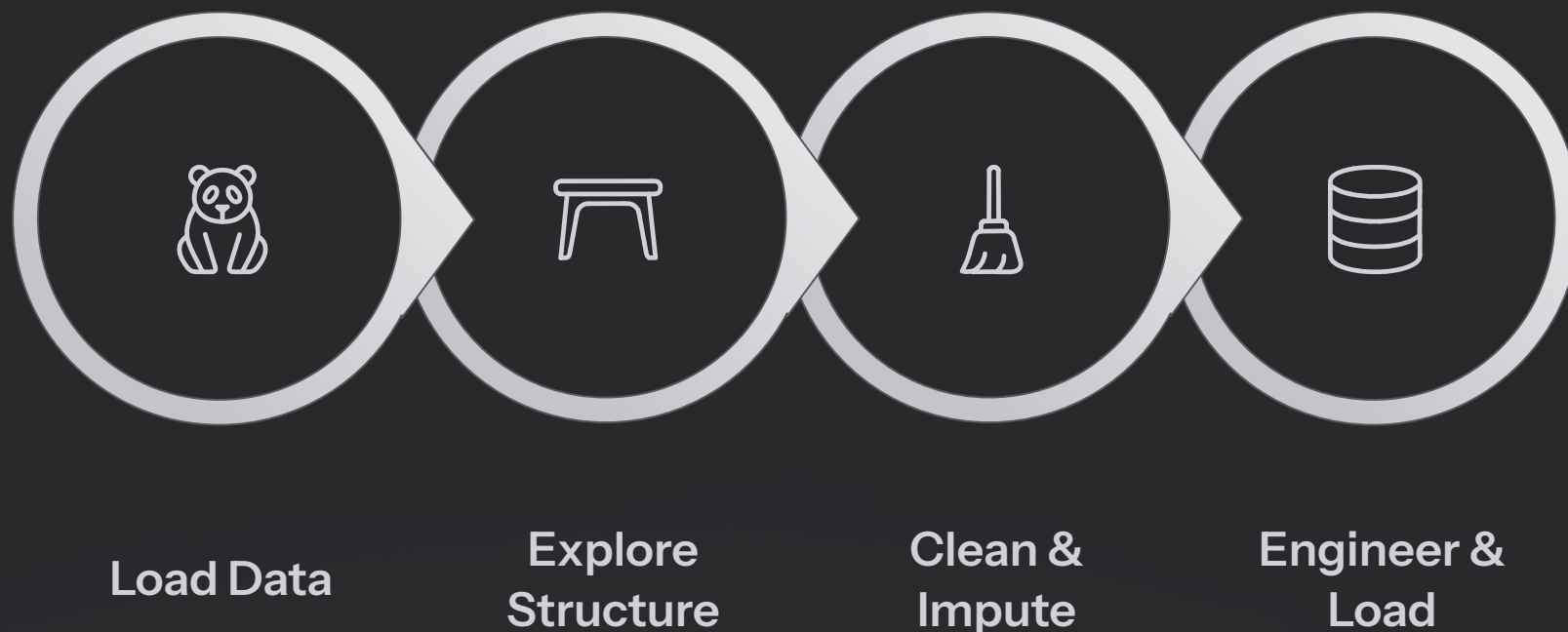
Purchase Details: Item Purchased, Category, Purchase Amount, Season, Size, and Color describe what customers are buying and under what conditions.

Shopping Behavior: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, and Shipping Type capture how customers interact with the purchasing process.

The **Review Rating** column had 37 missing values — less than 1% of the dataset — which were imputed using the median rating per product category during the data cleaning phase.

Exploratory Data Analysis in Python

The analysis began with data preparation and cleaning in Python using `pandas`. The dataset was imported and initial exploration was performed using `df.info()` to check structure and `.describe()` for summary statistics. The descriptive statistics revealed key patterns: the average customer age was **44.1 years** (std 15.2), ranging from 18 to 70; the mean purchase amount was **\$59.76** (std \$23.69), ranging from \$20 to \$100; and the average review rating was **3.75** (std 0.72), ranging from 2.5 to 5.0. Gender distribution was skewed toward male customers (2,652 of 3,900 records), and the most frequently purchased item was the Blouse (171 occurrences). Clothing was the dominant category with 1,737 purchases, and Montana was the most common location (96 records).



Data Cleaning & Standardization

Missing values in **Review Rating** were imputed using the median rating of each product category. All columns were renamed to **snake_case** for consistency and readability. A data consistency check verified whether `discount_applied` and `promo_code_used` were redundant — they were, and `promo_code_used` was dropped from the dataset.

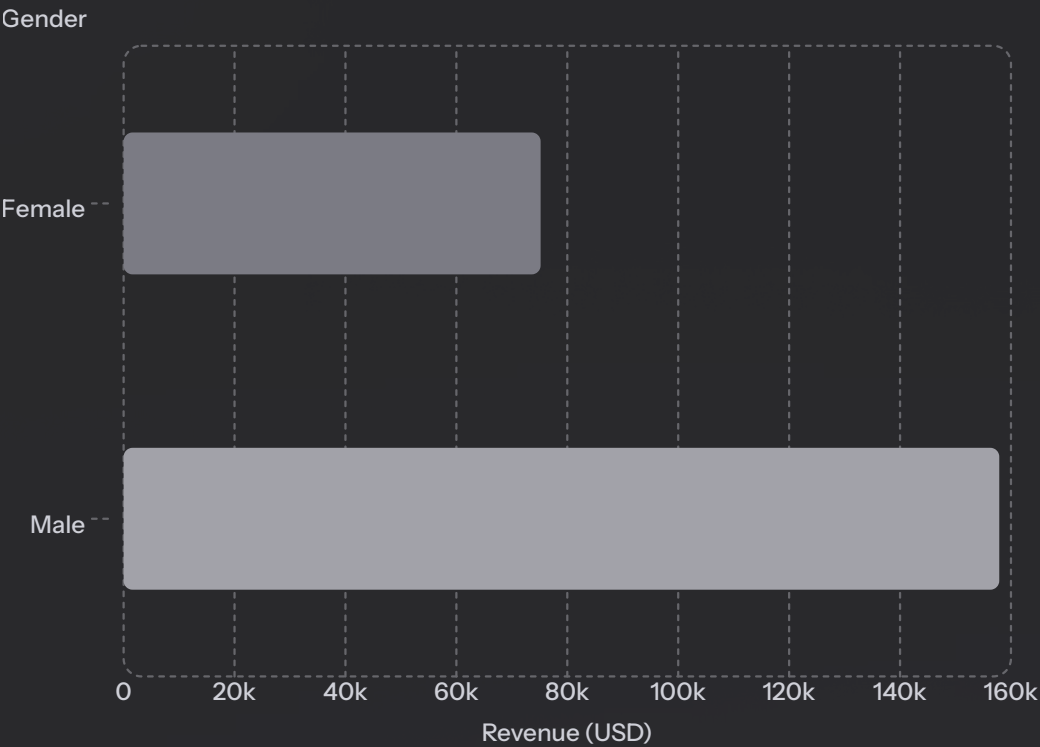
Feature Engineering

Two new features were created to enrich the analysis: an **age_group** column by binning customer ages into meaningful segments, and a **purchase_frequency_days** column derived from purchase data to quantify buying cadence. The cleaned DataFrame was then loaded into **PostgreSQL** for structured SQL analysis.

SQL Analysis: Revenue & Customer Spending

Structured analysis in PostgreSQL addressed key business questions around revenue distribution, discount behavior, and shipping preferences. The first query compared total revenue generated by male vs. female customers, revealing a significant gender-based spending gap. A second query identified customers who used discounts but still spent above the average purchase amount — a segment representing high-value, deal-seeking shoppers. A third query compared average purchase amounts between Standard and Express shipping to assess whether shipping type correlates with basket size.

Revenue by Gender

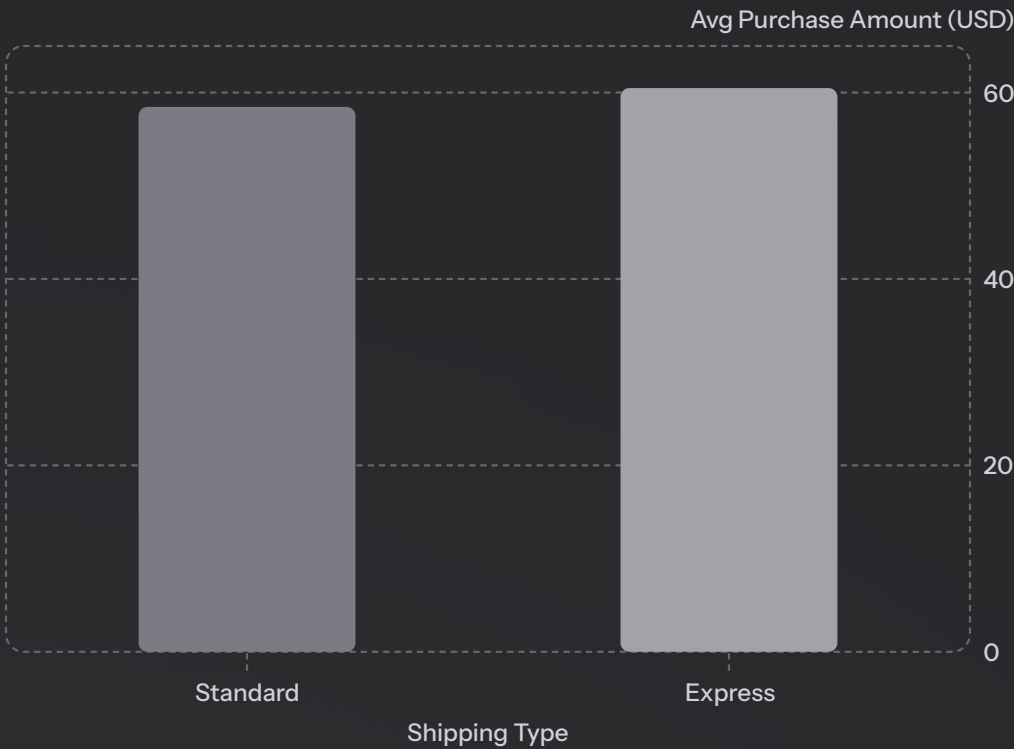


Male customers generated **\$157,890** in total revenue — more than double the **\$75,191** generated by female customers. This 2.1x gap highlights a critical opportunity to increase engagement and spending among female shoppers through targeted campaigns and product recommendations.

High-Spending Discount Users

A query identified customers who used discounts but still spent above the average purchase amount. This segment represents **839 customers** — deal-seekers who are already high-value buyers. These customers are prime candidates for loyalty incentives and personalized upsell offers, as they demonstrate both price sensitivity and strong purchasing power.

Shipping Type Comparison



Express shipping customers spend slightly more on average (**\$60.48**) compared to Standard shipping (**\$58.46**), suggesting that customers willing to pay for faster delivery may also have higher overall purchase intent.

SQL Analysis: Products, Ratings & Discounts

Further SQL queries examined product-level performance — identifying top-rated items, analyzing subscription-based spending patterns, and measuring discount dependency across products. These insights help inform product positioning, promotional strategy, and subscription program design.

Top 5 Products by Average Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Footwear and accessories dominate the top-rated list. Gloves lead at **3.86**, followed closely by Sandals (3.84) and Boots (3.82). These high-rated products should be prominently featured in marketing campaigns and recommendation engines.

Discount-Dependent Products

The following products had the highest percentage of discounted purchases, indicating strong price sensitivity among their buyers:

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Hats are the most discount-dependent product, with **50% of purchases** made with a discount applied. Sneakers, Coats, Sweaters, and Pants all show discount rates above 47%, suggesting that promotional pricing is a key driver for these categories. This insight is critical for margin management — while discounts drive volume, they may erode profitability if not carefully controlled.

SQL Analysis: Subscribers, Segments & Age Groups

Three additional SQL queries examined subscription behavior, customer segmentation, and revenue contribution by age group — providing a demographic and behavioral profile of the customer base.

Subscribers vs. Non-Subscribers

Status	Customers	Avg Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Non-subscribers represent **73% of customers** and generate **\$170,436** in revenue — nearly 3x the subscriber revenue of \$62,645. However, average spend per transaction is nearly identical (\$59.49 vs. \$59.87), indicating that the revenue gap is driven by subscriber volume, not basket size. Converting non-subscribers is a major growth lever.

Customer Segmentation

Customers were classified into three segments based on purchase history:

Loyal

3,116 customers — the dominant segment, representing 80% of the base

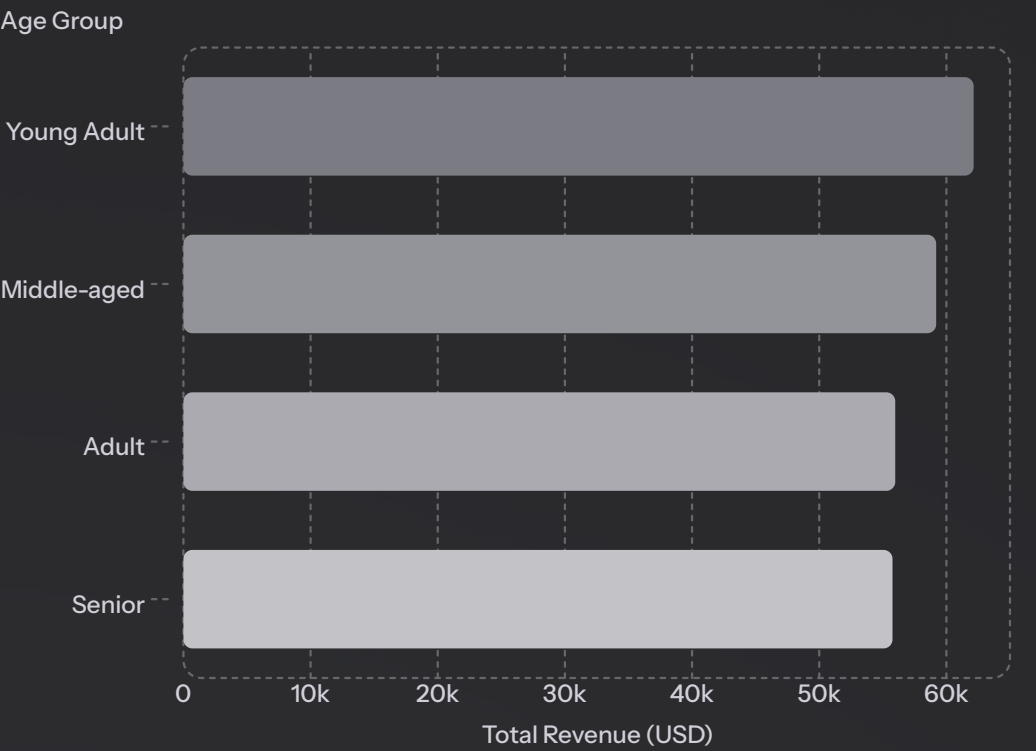
Returning

701 customers — repeat buyers with growth potential

New

83 customers — first-time buyers requiring onboarding

Revenue by Age Group



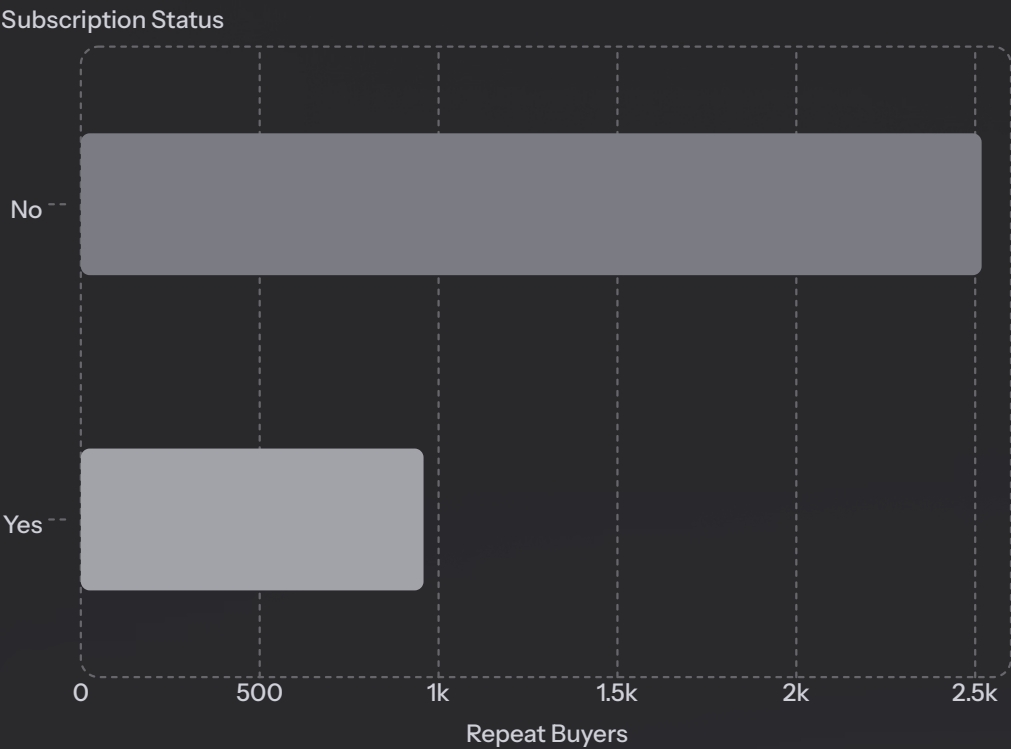
Young Adults generate the highest total revenue at **\$62,143**, followed by Middle-aged (\$59,197), Adult (\$55,978), and Senior (\$55,763). The relatively even distribution across age groups suggests broad product appeal, with a slight edge toward younger demographics.

SQL Analysis: Repeat Buyers & Top Products by Category

Two final SQL analyses examined the relationship between repeat purchasing behavior and subscription status, and identified the top-selling products within each category to inform inventory and marketing priorities.

Repeat Buyers & Subscriptions

A query checked whether customers with more than 5 purchases are more likely to subscribe. The results show that among repeat buyers, **2,518 are non-subscribers** while only **958 are subscribers**. This indicates that the majority of high-frequency buyers have not yet converted to subscriptions — a significant untapped opportunity for subscription program promotion targeted at this engaged segment.



Top 3 Products per Category

The most purchased products within each category were identified to guide product positioning and campaign focus:

Category	Rank	Product	Orders
Accessories	1	Jewelry	171
Accessories	2	Sunglasses	161
Accessories	3	Belt	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Clothing	3	Shirt	169
Footwear	1	Sandals	160
Footwear	2	Shoes	150
Footwear	3	Sneakers	145
Outerwear	1	Jacket	163
Outerwear	2	Coat	161

Clothing dominates order volume, with Blouse, Pants, and Shirt each generating 169–171 orders. Jewelry leads Accessories (171 orders), Sandals lead Footwear (160 orders), and Jacket leads Outerwear (163 orders). These top performers should be prioritized in inventory planning and featured prominently in marketing campaigns.

Power BI Dashboard: Key Metrics

An interactive **Power BI dashboard** titled "Customer Behavior Dashboard" was built to present insights visually. The dashboard includes a sidebar with filters for Subscription Status, Gender, Category, and Shipping Type, enabling stakeholders to slice and explore the data dynamically. The main area features key performance metrics and a series of charts providing a comprehensive overview of customer data across multiple dimensions.

3.9K

Total Customers

Across all transaction records in the dataset

\$59.76

Avg Purchase

Mean transaction value across all categories

3.75

Avg Rating

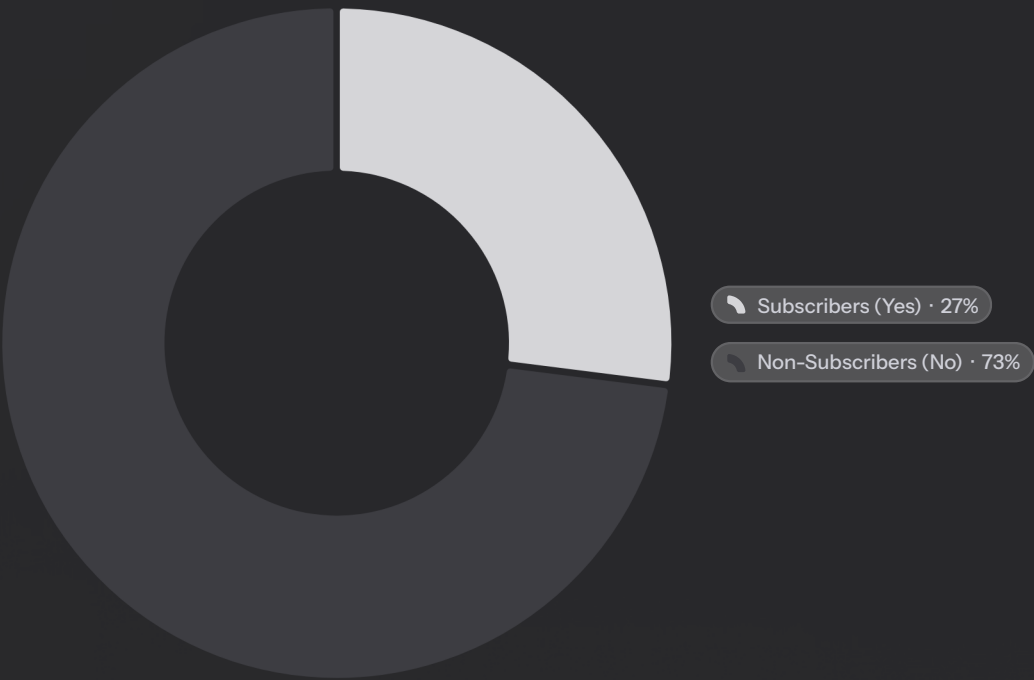
Mean customer review rating out of 5.0

These three headline metrics provide an at-a-glance summary of the customer base size, spending behavior, and satisfaction level. The dashboard's filter panel allows users to drill into these metrics by Subscription Status, Gender, Category, and Shipping Type for deeper segment-level analysis.

Power BI Dashboard: Visual Charts

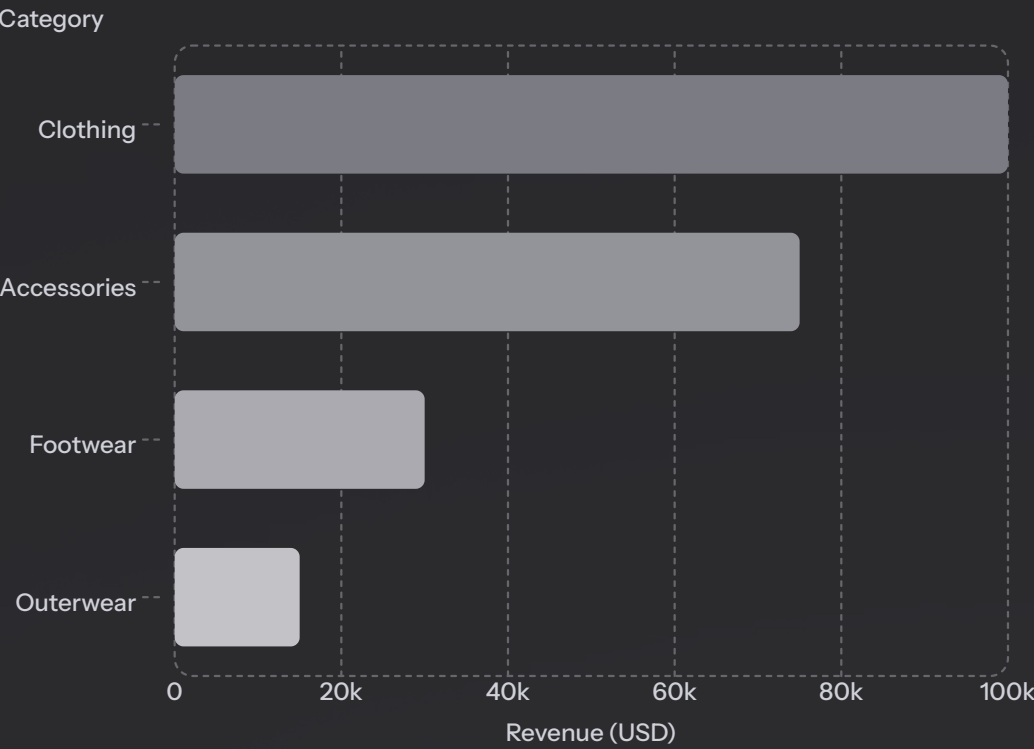
Beyond the headline metrics, the dashboard includes six charts that break down performance by subscription status, product category, and age group — enabling stakeholders to identify trends and opportunities at a glance.

% Customers by Subscription Status



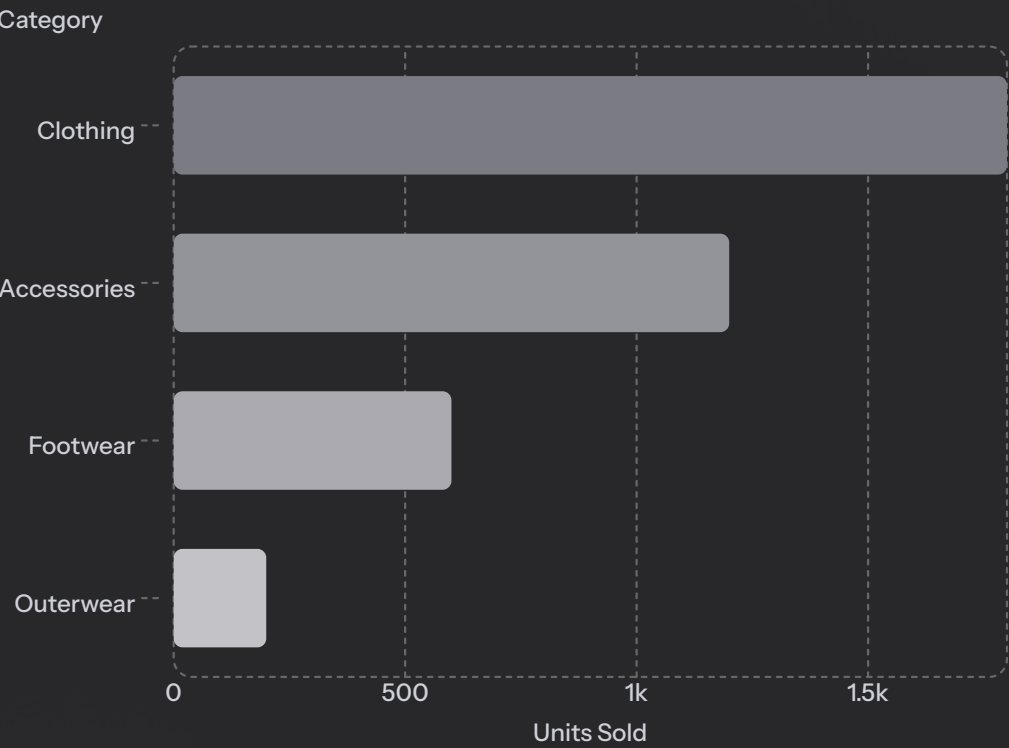
Only **27%** of customers are subscribers, while **73%** remain non-subscribers — confirming the massive conversion opportunity identified in the SQL analysis.

Revenue by Category



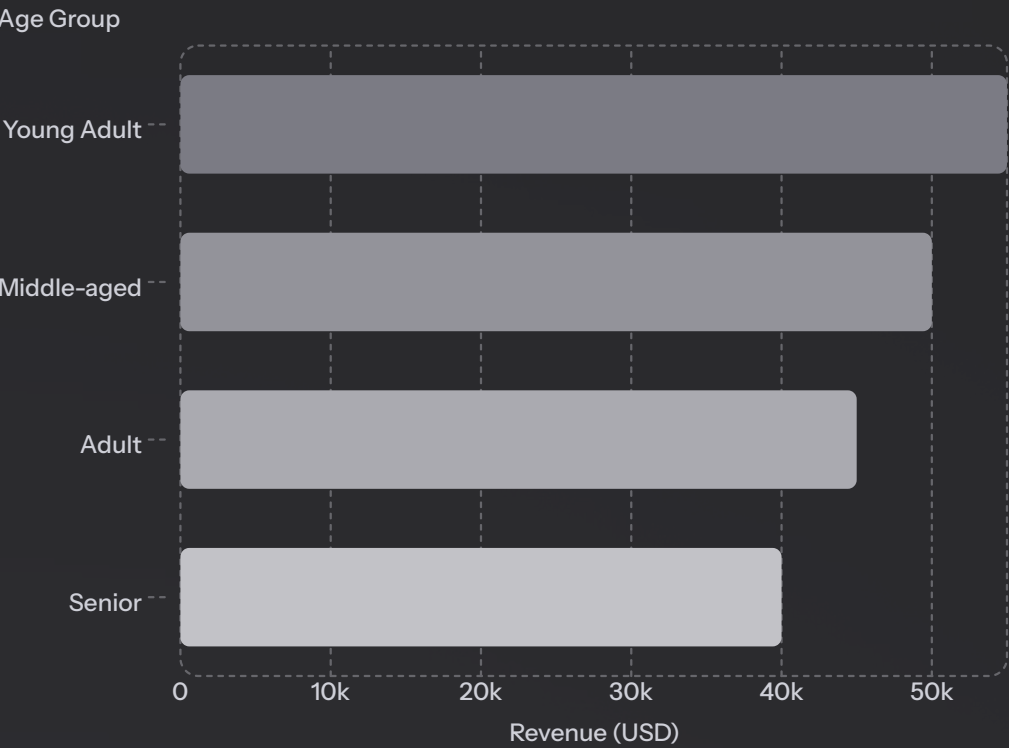
Clothing dominates revenue at **\$100K**, followed by Accessories (\$75K), Footwear (\$30K), and Outerwear (\$15K). Clothing and Accessories together account for the vast majority of total revenue.

Sales by Category



Clothing leads in unit sales with **1,800** units, followed by Accessories (1,200), Footwear (600), and Outerwear (200). The sales volume pattern mirrors the revenue distribution, confirming Clothing as the core business driver.

Revenue by Age Group



Young Adults lead in revenue contribution at **\$55K**, followed by Middle-aged (\$50K), Adult (\$45K), and Senior (\$40K). The dashboard visualizations confirm that younger demographics are the highest-value segment, while revenue is relatively well-distributed across all age groups.

Business Recommendations

Based on the analytical findings, five strategic recommendations are proposed to drive revenue growth, improve customer retention, and optimize marketing efficiency.

Boost Subscriptions

With only **27%** of customers subscribed and non-subscribers generating 3x more total revenue, promoting exclusive subscriber benefits is the highest-impact opportunity. Target the **958 repeat buyers** without subscriptions and the broader **2,518 non-subscribing repeat buyers** with personalized upgrade offers.

Customer Loyalty Programs

The dataset shows **3,116 Loyal** customers, **701 Returning**, and only **83 New** customers. Reward repeat buyers to move Returning customers into the Loyal segment, and design onboarding incentives to convert New customers into Returning buyers.

Review Discount Policy

Products like Hats (**50% discount rate**), Sneakers (49.66%), and Coats (49.07%) show heavy discount dependency. Balance sales volume boosts with margin control — consider shifting from blanket discounts to targeted, behavior-triggered offers for the **839 high-spending discount users** identified in the analysis.

Product Positioning

Highlight top-rated products — Gloves (3.86★), Sandals (3.84★), Boots (3.82★) — and best-sellers like Blouse, Pants, and Jewelry in marketing campaigns. Clothing's **\$100K revenue** and 1,800 units sold confirm it as the flagship category for promotional investment.

Targeted Marketing

Focus efforts on **Young Adult** and **Middle-aged** segments, which generate the highest revenue. Prioritize Express Shipping users who demonstrate higher average spend (**\$60.48** vs. \$58.46). Address the male revenue dominance (**\$157,890** vs. **\$75,191**) with campaigns designed to increase female customer engagement and spending.